Introduction

We are told today that we are living in an age of massive transformation. Terms like the sharing economy, the gig economy, and the fourth industrial revolution are tossed around, with enticing images of entrepreneurial spirit and flexibility bandied about. As workers, we are to be liberated from the constraints of a permanent career and given the opportunity to make our own way by selling whatever goods and services we might like to offer. As consumers, we are presented with a cornucopia of on-demand services and with the promise of a network of connected devices that cater to our every whim. This is a book on this contemporary moment and its avatars in emerging technologies: platforms, big data, additive manufacturing, advanced robotics, machine learning, and the internet of things. It is not the first book to look at these topics, but it takes a different approach from others. In the existing literature, one group of commentaries focuses on the politics of emerging technology, emphasising privacy and state surveillance but leaving aside economic issues around ownership and profitability. Another group looks at how corporations are embodiments of particular ideas and values and criticises them for not acting humanely – but, again, it neglects the economic context and the imperatives of a capitalist system.[1] Other scholars do examine these emerging economic trends but present them as sui generis phenomena, disconnected from their history. They never ask why we have this economy today, nor do they recognise how today's economy responds to yesterday's problems. Finally, a number of analyses report on how poor the smart economy is for workers and how digital labour represents a shift in the relationship between workers and capital, but they leave aside any analysis of broader economic trends and intercapitalist competition.[2]

기존연구: 정치에 초점,
△ 경제적 측면 간과

The present book aims to supplement these other perspectives by giving an economic history of capitalism and digital technology, while recognising the diversity of economic forms and the competitive tensions inherent in the contemporary economy. The simple wager of the book is that we can learn a lot about major tech companies by taking them to be economic actors within a capitalist mode of production. This means abstracting from them as cultural actors defined by the values of the Californian ideology, or as political actors seeking to wield power. By contrast, these actors are compelled to seek out profits in order to fend off competition. This places strict limits on what constitutes possible and predictable expectations of what is likely to occur. Most notably, capitalism demands that firms constantly seek out new avenues for profit, new markets, new commodities, and new means of exploitation.

경제와 결합시켜 분석

For some, this focus on capital rather than labour may suggest a vulgar economism; but, in a world where the labour movement has been significantly weakened, giving capital a priority of agency seems only to reflect reality.

Where, then, do we focus our attention if we wish to see the effects of digital technology on capitalism? We might turn to the technology sector,[3] but, strictly speaking, this sector remains a relatively small part of the economy. In the United States it currently contributes around 6.8 per cent of the value added from private companies and employs about 2.5 per cent of the labour force.[4] By comparison, manufacturing in the deindustrialised United States employs four times as many people. In the United Kingdom manufacturing employs nearly three times as many people as the tech sector.[5] This is in part because tech companies are notoriously small. Google has around 60,000 direct employees, Facebook has 12,000, while WhatsApp had 55 employees when it was sold to Facebook for $19 billion and Instagram had 13 when it was purchased for $1 billion. By comparison, in 1962 the most significant companies employed far larger numbers of workers: AT&T had 564,000 employees, Exxon had 150,000 workers, and GM had 605,000 employees.[6] Thus, when we discuss the digital economy, we should bear in mind that it is something broader than just the tech sector defined according to standard classifications.

As a preliminary definition, we can say that the digital economy refers to those businesses that increasingly rely upon information technology, data, and the internet for their business models. This is an area that cuts across traditional sectors – including manufacturing, services, transportation, mining, and telecommunications – and is in fact becoming essential to much of the economy today. Understood in this way, the digital economy is far more important than a simple sectoral analysis might suggest. In the first place, it appears to be the most dynamic sector of the contemporary economy – an area from which constant innovation is purportedly emerging and that seems to be guiding economic growth forward. The digital economy appears to be a leading light in an otherwise rather stagnant economic context. Secondly, digital technology is becoming systematically important, much in the same way as finance. As the digital economy is an increasingly pervasive infrastructure for the contemporary economy, its collapse would be economically devastating. Lastly, because of its dynamism, the digital economy is presented as an ideal that can legitimate contemporary capitalism more broadly. The digital economy is becoming a hegemonic model: cities are to become smart, businesses must be disruptive, workers are to become flexible, and governments must be lean and intelligent. In this environment those who work hard can take advantage of the changes and win out. Or so

제조업 수익성 하락 → 생산 침체…
이를 벗어나기 위해 '데이터'를 수단으로 사용하게 됨
⇒ '플랫폼' 등장 + 독점기업 부상

we are told.

The argument of this book is that, with a long decline in manufacturing profitability, capitalism has turned to data as one way to maintain economic growth and vitality in the face of a sluggish production sector. In the twenty-first century, on the basis of changes in digital technologies, data have become increasingly central to firms and their relations with workers, customers, and other capitalists. The platform has emerged as a new business model, capable of extracting and controlling immense amounts of data, and with this shift we have seen the rise of large monopolistic firms. Today the capitalism of the high- and middle-income economies is increasingly dominated by these firms, and the dynamics outlined in this book suggest that the trend is only going to continue. The aim here is to set these platforms in the context of a larger economic history, understand them as means to generate profit, and outline some of the tendencies they produce as a result.

In part, this book is a synthesis of existing work. The discussion in Chapter 1 should be familiar to economic historians, as it outlines the various crises that have laid the groundwork for today's post-2008 economy. It attempts to historicise emerging technologies as an outcome of deeper capitalist tendencies, showing how they are implicated within a system of exploitation, exclusion, and competition. The material in Chapter 2 should be fairly well known to those who follow the business of technology. In many ways, the chapter is an attempt to give clarity to various ongoing discussions in that world, as it lays out a typology and genesis of platforms. By contrast, Chapter 3 hopefully offers something new to everyone. On the basis of the preceding chapters, it attempts to draw out some likely tendencies and to make some broad-brush predictions about the future of platform capitalism. These forward-looking prognoses are essential to any political project. How we conceptualise the past and the future is important for how we think strategically and develop political tactics to transform society today. In short, it makes a difference whether we see emerging technologies as inaugurating a new regime of accumulation or as continuing earlier regimes. This has consequences on the possibility of a crisis and on deciding where that crisis might emerge from; and it has consequences on our envisaging the likely future of labour under capitalism. Part of the argument of this book is that the apparent novelties of the situation obscure the persistence of longer term trends, but also that today presents important changes that must be grasped by a twenty-first-century left. Understanding our position in a broader context is the first step to creating strategies for transforming it.

Notes

3
Great Platform Wars

If platforms are the emerging business model for the digital economy, how do they appear when set in the longer history of capitalism? In particular, up to this point we have largely left out one of the fundamental drivers of capitalism: intracapitalist competition. In Chapter 1 we set out the context of the long downturn – that period since the 1970s when the global economy has been saddled by overcapacity and overproduction in the manufacturing sector. As companies were unwilling and unable to destroy their fixed capital or to invest in new lines, international competition has steadily continued and, alongside it, the crisis of overcapacity in manufacturing. Unable to generate growth in this situation, in the 1990s the United States began trying to stimulate the economy through an asset-price Keynesianism that operated by inducing low interest rates in order to generate higher asset prices and a wealth effect that would spark broader economic growth. This led to the dot-com boom of the 1990s and to the housing bubble of the early years of the twenty-first century. Today, as we saw in the previous chapter, asset-price Keynesianism continues apace and is one of the fundamental drivers behind the current mania for tech start-ups. Yet, behind the shiny new technology and slick façade of app interfaces, what broader consequences do these new firms hold for capitalism? In this chapter we will step back to look at the tendencies unleashed by these new firms into the broader economic environment of the long downturn. Some argue that capitalism renews itself through the creation and adoption of new technological complexes: steam and railways, steel and heavy engineering, automobiles and petrochemicals – and now information and communications technologies.[1] Are we witnessing the adoption of a new infrastructure that might revive capitalism's moribund growth? Will competition survive in the digital era, or are we headed for a new monopoly capitalism?

: 네트워크 효과 → 독점화 영향 내재 (최지의 우위나 지속적인 지배3 이어짐) ex) 미국: 페이스북과 구글이 온라인 광고 수익의 76% 차지

With network effects, a tendency towards monopolisation is built into the DNA of platforms: the more numerous the users who interact on a platform, the more valuable the entire platform becomes for each one of them. Network effects, moreover, tend to mean that early advantages become solidified as permanent positions of industry leadership. Platforms also have a unique ability to link together and consolidate multiple network effects. Uber, for instance, benefits from the network effects of more and more drivers as well as from the network effects of more and more riders.[2] Leading platforms tend

consciously to perpetuate themselves in other ways as well. Advantages in data collection mean that the more activities a firm has access to, the more data it can extract and the more value it can generate from those data, and therefore the more activities it can gain access to. Equally, access to a multitude of data from different areas of our life makes prediction more useful, and this stimulates the centralisation of data within one platform. We give Google access to our email, our calendars, our video histories, our search histories, our locations – and, with each aspect provided to Google, we get better predictive services as a result. Likewise, platforms aim to facilitate complementary products: useful software built for Android leads more users to use Android, which leads more developers to develop for Android, and so on, in a virtuous circle. Platforms also seek to build up ecosystems of goods and services that close off competitors: apps that only work with Android, services that require Facebook logins. All these dynamics turn platforms into monopolies with centralised control over increasingly vast numbers of users and the data they generate. We can get a sense of how significant these monopolies already are by looking at how they consolidate ad revenue: in 2016 Facebook, Google, and Alibaba alone will take half of the world's digital advertising.[3] In the United States, Facebook and Google receive 76 per cent of online advertising revenue and are taking 85 per cent of every new advertising dollar.[4]

Yet it is also true that capitalism develops not only greater means for monopoly but also greater means for competition. The emergence of the corporation form, the rise of large financial institutions, and the monetary resources behind states all point to its capacity to initiate new lines of industry and to topple existing monopolies.[5] Equally importantly, digital platforms tend to arise in industries that are subject to disruption by new competitors.[6] Monopolies, in this view, should only ever be temporary. The challenge today, however, is that capital investment is not sufficient to overturn monopolies; access to data, network effects, and path dependency place even higher hurdles in the way of overcoming a monopoly like Google. This does not mean the end of competition or of the struggle for market power, but it means a change in the form of competition.[7] In particular, this is a shift away from competition over prices (e.g. many services are offered for free). Here we come to an essential point. Unlike in manufacturing, in platforms competitiveness is not judged solely by the criterion of a maximal difference between costs and prices; data collection and analysis also contribute to how competitiveness is judged and ranked. This means that, if these platforms wish to remain competitive, they must intensify their extraction, analysis, and control of data – and they must invest in the fixed capital to do so. And while

디지털 플랫폼 산업
: 새로운 경쟁자에 의해 교란可
→ 독점은 언제나 일시적이다
△, 엄청난 만큼 자본투입만으로
독점을 무너뜨릴수 없다

플랫폼 산업: 경쟁력이 단지
비용·가격 차원에서 형성X,
데이터 수집·분석이 중요하다!
(이를 위한 고정 인프라 투자중요)

their genetic drive is towards monopolisation, at present they are faced with an increasingly competitive environment comprised of other great platforms.

Tendencies

Since platforms are grounded upon the extraction of data and the generation of network effects, certain tendencies emerge from the competitive dynamics of these large platforms: expansion of extraction, positioning as a gatekeeper, convergence of markets, and enclosure of ecosystems. These tendencies then go on to be installed in our economic systems.

At one level, the expansion of platforms is driven by the cross-subsidisation of services used to draw users into a network. If a service appears likely to draw consumers or suppliers into the platform, then a company may develop the tools to do so. Yet expansion is also driven by factors other than user demand. One such factor is the drive for further data extraction. If collecting and analysing this raw material is the primary revenue source for these companies and gives them competitive advantages, there is an imperative to collect more and more. As one report notes, echoing colonialist ventures: 'From a data-production perspective, activities are like lands waiting to be discovered. Whoever gets there first and holds them gets their resources – in this case, their data riches.'[8] For many of these platforms, the quality of the data is of less interest than their quantity and diversity.[9] Every action performed by a user, no matter how minute, is useful for reconfiguring algorithms and optimising processes. Such is the importance of data that many companies could make all of their software open-source and still maintain their dominant position due to their data.[10] Unsurprisingly, then, these companies have been prolific purchasers and developers of assets that enable them to expand their capacity for gaining information. Mergers relating to big data, for instance, have doubled between 2008 and 2013.[11] Their vast cash glut and frequent use of tax havens contributed to making this possible. A large surplus of capital sitting idle has enabled these companies to build and expand an infrastructure of data extraction.

많은 데이터를 얻기 위해
인프라 구축·투자↑

This is the context in which we should understand the significant investments made in the consumer internet of things (IoT), where sensors are placed in consumer goods and homes.[12] For example, Google's investment in Nest, a heating system for residential homes, makes much more sense when it is understood as the extension of data extraction. The same goes for Amazon's new device, Echo, an always-on device that consumers place in their homes. At the mention of its name, Echo will respond to questions; but it is also capable of recording activities around it. It is not difficult to see how this

might be useful for a company trying to understand consumer preferences. Similar devices already exist in many phones – Siri for Apple, Google Now for Android, not to mention the emergence of smart TVs.[13] Wearable technologies are another major element of consumer IoT. Nike, for instance, is using wearables and fitness technology to bring users onto its platform and extract their data. While all these devices may have some use value for consumers, the field has not been driven by consumers clamouring for them. Instead, consumer IoT is only fully intelligible as a platform-driven extension of data recording into everyday activities. With consumer IoT, our everyday behaviours start to be recorded: how we drive, how many steps we take, how active we are, what we say, where we go, and so on. This is simply an expression of an innate tendency within platforms. It is therefore no surprise that one of Facebook's most recent acquisitions, the Oculus Rift VR system, is able to collect all sorts of data on its users and uses this information as part of the sales pitch to advertisers.[14]

IoT는 플랫폼이 데이터수집 범위를 일상에까지 확장하려는 전략

The fact that the information platform requires an extension of sensors means that it is countering the tendency towards a lean platform. These are not asset-less companies – far from it; they spend billions of dollars to purchase fixed capital and take other companies over. Importantly, 'once we understand this [tendency], it becomes clear that demanding privacy from surveillance capitalists or lobbying for an end to commercial surveillance on the Internet is like asking Henry Ford to make each Model T by hand'.[15] Calls for privacy miss how the suppression of privacy is at the heart of this business model. This tendency involves constantly pressing against the limits of what is socially and legally acceptable in terms of data collection. For the most part, the strategy has been to collect data, then apologise and roll back programs if there is an uproar, rather than consulting with users beforehand.[16] This is why we will continue to see frequent uproars over the collection of data by these companies.

프라이버시의 억압 = 플랫폼 모델의 핵심 동력

If data collection is a key task of platforms, analysis is the necessary correlate. The proliferation of data-generating devices creates a vast new repository of data, which requires increasingly large and sophisticated storage and analysis tools, further driving the centralisation of these platforms.[17] If expanding the capacity to collect data is one competitive imperative for these companies, developing corresponding means of analysis is another. Advances in hardware, database organisation, and network infrastructure therefore all play significant roles in gaining speed and insight advantages over one's competitors. Much of Google's initial success, for instance, stemmed from its pioneering work of creating useful internal software and innovative hardware architecture.[18] Rather uniquely, Google designs and builds its own custom

servers rather than purchasing standard servers off the market – again, in an effort to gain competitive advantage.[19] And, while often it eventually releases information about its operations (which have then been copied by numerous others), it only does so after it has gained a clear advantage.[20] It is the importance of analytics that lets us understand why Google is heavily invested in artificial intelligence (AI) research as well, given that this is the key area for developing a competitive advantage over other platforms. Google is the biggest investor in this area, but Amazon, Salesforce, Facebook and Microsoft are all investing heavily in AI as well. Firms also have imperatives to develop the entire stack, and not just one area of it (e.g. data management, or analytical tools).[21] Bottlenecks in the flow of data from sensors to commodity are an impediment to producing more value. The result is a tendency to increasingly take on all the features of the stack, from hardware to software.

전체 생태계를 장악하는 방식으로 경쟁

This is matched by a second tendency, whereby expansion across the ecosystem around a core business segment is driven in part by the need to occupy key positions within the ecosystem. These evade traditional distinctions: they are neither horizontal mergers (combining companies that directly compete), nor vertical mergers (combining companies within the same supply chain), nor conglomerate mergers (combining suppliers of similar and complementary products).[22] These mergers consist not so much in the vertical integration of classic Fordist firms or in the lean competencies of the post-Fordist era; they are more like rhizomatic connections driven by a permanent effort to place themselves in key platform positions. Let us take a first example. As access to the internet shifted away from desktop computing and towards handheld smartphones, control over the operating system (OS) platforms became essential. The shift caused companies to rush and implant themselves into the smartphone market: Google followed in Apple's footsteps, and Amazon and Facebook later attempted to catch up. Google used the traditional platform tactic of cross-subsidisation in order to occupy the mobile OS market: it licensed Android for free to hardware makers, so as to undercut Apple's enclosed system. The gambit worked, and Android today has more than 80 per cent of the market and is the most widely used OS on any device.[23] Similar competitive battles – and subsequent business expansions – have been going on at the interface level as well. As the primary means by which users interact with platforms, interfaces occupy a key intermediary position in the broader ecosystem. For the last decade, Google's search engine has been the primary interface into the rest of the internet, outpacing any other effort. Rival platforms have had to route around Google's search engine dominance by extending their business into new interface areas.

One expression of this is that search engines within apps (rather than the open web) are becoming increasingly widespread. Instead of searching the internet through Google, users can search internally, on Amazon or Facebook. If people move into apps or start searching on Amazon instead of Google, these are threats to Google's basic business model.

Every major platform company is increasingly positioning itself in the natural language interface market as well. In 2016 Facebook began a major push for 'chatbots' – that is, low-level AI programmes that would converse with users on Facebook's platform. (This is also why Facebook – and numerous other companies – are investing heavily in AI and the natural language processing needed to enable chatbots.) The bet is that these chatbots will become the preferred way for users to interact with the internet. On this open platform, businesses would be given the tools to develop their own bots and create intuitive means for users to order food, buy a train ticket, or make a dinner reservation.[24] Rather than using a separate app or website for accessing businesses and services, users would simply access them through Facebook's platform, which would make Facebook's chatbot platform the primary interface for commercial transactions online. Rather than trying to compete against Google's search engine or Amazon's logistics network, Facebook is trying to dominate the e-commerce platform by dominating the interface.

Whether this will work or not is debatable, but the principle is that these companies expand in intelligible ways, namely in order to take over key positions. Similar principles hold for Apple's, Google's, and Facebook's efforts to become a payment platform and to build the base for making economic transactions by collecting a small fee in every case, along with the data. This is also true of competition over mapping: Uber's bid to purchase a mapping provider, Google's use of Google Maps cars for its navigational basis, Apple's construction of its own locational services in 2012, and Uber's potential for building its own proprietary mapping provider. The aim is to occupy a position in the stack; where certain levels are more foundational and therefore more powerful, but also more difficult to secure and more subject to monopoly power and to major entry barriers. While we might think that being lower in the stack is correlated with greater power, this is not necessarily the case. Perhaps surprisingly, network providers (i.e. those that provide the basic telecommunications infrastructure) are in a low margin position in the ecosystem around platforms – a position that has compelled them to push for discriminatory pricing in moving data around (the end of 'net neutrality') as a way to generate more revenues.[25] The strategic importance of a position has much more to do with controlling data from businesses and customers than with just being lower in the stack.

These first two expansionary tendencies give platform monopolies a distinct path of expansion by comparison to traditional business models premised on vertical integration, horizontal integration, or conglomeration.[26] Instead, platform expansion is driven by the need for more data, which leads to what we might call the convergence thesis: the tendency for different platform companies to become increasingly similar as they encroach upon the same market and data areas. Currently there is a plethora of different platform models that have emerged from contingent economic conditions and strategic decisions based upon strengths in different areas.[27] One key question is what the future development of these forms holds: Will they converge into an ur-platform model? Or will they diverge and maintain competitiveness through specialisation? Given the need to expand data extraction and to position oneself in strategic locations, it would appear that companies are tendentially drawn into similar areas. This means that, despite their differences, companies like Facebook, Google, Microsoft, Amazon, Alibaba, Uber, and General Electric (GE) are also direct competitors. IBM, for instance, has moved into the platform business, purchasing Softlayer for cloud computing, and BlueMix for software development. The convergence thesis helps explain why Google is lobbying with Uber on self-driving cars and why Amazon and Microsoft have been discussing partnerships with German automakers on the cloud platform required by self-driving cars.[28] Alibaba and Apple have made major investments in Didi, Apple's partnership being particularly strategic, given that iPhones are the major interface to taxi services. And nearly all of the major platforms are working to develop medical data platforms. The trend to convergence is igniting international competition as well: intense struggles occur in India and China over who will dominate the ride-sharing industry (Uber, Didi, Lyft) and who will dominate e-commerce (Amazon, Alibaba, Flipkart). Alibaba is already the largest e-commerce site in the world as measured by the volume of its sales,[29] and Flipkart is valued at around $15 billion. Under the pressures of competition and the subsequent imperative to expand, we should expect these platforms to acquire as many companies as they need. Even second-tier platforms like Twitter and Yahoo are potential purchases, given the vast cash glut being held by the top tier of platforms (indeed, as I wrote this book, Microsoft purchased LinkedIn for $26 billion, gaining access to data on the changing interests, skills, and jobs of millions of workers). By 2015, global mergers and acquisitions had shot up more than 40 per cent above pre-crisis levels,[30] and the leading platforms have all made major moves in acquiring the resources needed to compete with their rivals. Ultimately, we see convergence – and therefore competition – across the field: smartphones, e-book readers, consumer IoT, cloud platforms, videochat services, payment services, driverless cars, drones, virtual reality, social

서로 다른 플랫폼 기업들이 동일한 시장과 데이터 영역에 진입하면서 점차 유사해짐… 수렴

networking, interfaces, network provision, search, and probably much more in the future.

A third dominant tendency is the funnelling of data extraction into siloed platforms. When extensive means are not sufficient for competitive advantage, this approach tries to tie users and data to the platform by locking them in through various measures: dependency on a service, inability to use alternatives, or lack of data portability, for instance. Apple is perhaps the leader in this tendency, as it makes its services and devices all highly interdependent and closed off to alternatives (with the notable exception of the semi-open App Store). Facebook is another clear example of this trend. Indeed, a major reason for Facebook's success is that, while Google dominated the open web through its search technology, Facebook was built as a closed platform that escaped the grips of Google. The aim for Facebook is to make it so that users never have to leave their enclosed ecosystem: news stories, videos, audio, messaging, email, and even buying consumer goods have all been progressively folded back into the platform itself. Enclosure takes on an even stricter form with Facebook's attempt to bring internet access to India and other countries through its Free Basics program. Facebook's own services would be provided for free, but other services would have to partner with Facebook and go through its platform, effectively enclosing the entirety of the internet into Mark Zuckerberg's silo.[31] While rejected in India, the Free Basics service is now active in 37 countries and used by over 25 million people.[32] Uber is also effectively building up a system that funnels passengers into its system. The decreased demand for non-Uber cabs means a decreased supply of non-Uber drivers, as more and more of the services move onto Uber. As more passengers turn to Uber's platform, non-Uber cab drivers will lose out and be forced onto Uber's platform if they are to survive. The same holds for passengers: as fewer non-Uber cabs roam the streets, the only way to guarantee a cab will eventually be through Uber's platform. The field of industrial platform is also almost certain to resolve into a series of enclosed spaces, as Siemens and GE are unable (and unwilling) to communicate with each other. Manufacturers will be locked into whichever ecosystem they choose. This is particularly important in terms of intracapitalist competition: as non-platform companies are forced to use platforms to continue their business, a divide will grow between these two groups. Non-platform companies will put pressure on platforms to lower their prices, and platforms will fight back by making switching platforms increasingly costly and monopolistic. Amazon, too, aims to be a closed platform, separated from Google. Rather than turning to an internet search engine for buying goods online, users would search for goods, compare, purchase, track, and review, all without ever leaving the Amazon

platform.

We also see the platform model driving the move from an open web to increasingly closed apps. The expansion of smartphones has led to more and more users interacting with the internet through apps rather than by visiting websites, and this is a way in which companies can both expand and close off data collection. As more users head into an app, those data are extracted there, while other platforms lose out. This trend also means that rivals seek to unmoor themselves from dependency on others: Dropbox is spending large amounts of money to separate itself from AWS, and Uber is seeking to untie itself from dependency on Google Maps. Even deeper down the stack, platforms are at work building their own network infrastructure. Google, for instance, has been building its own privatised internet – browsers, OSs, fibre networks, and data centres – where information may never have to journey across public infrastructure.[33] Likewise, Amazon's cloud network is nothing if not a private internet, and Microsoft and Facebook are collaborating to build their own transatlantic fibre cable.[34] Taken to a logical conclusion, this trend may lead to specialised platforms that give up on the idea of general computing and instead focus on optimising their particular services and the associated rents that go with these services.[35] In the end, the tendency of major platforms to grow to immense size thanks to network effects, combined with the tendency to converge towards a similar form, as market pressures dictate, leads them to use enclosure as a key means of competing against their rivals. If this analysis is right, then capitalist competition is driving the internet to fragment. There is no necessity to this outcome, as political efforts can stall or reverse it; but within a capitalist mode of production there are strong competitive pressures towards this end.

네트워크 효과 + 수렴
→ enclosure (폐쇄)

Challenges

For all the rhetoric of having overcome capitalism and of transitioning to a new mode of production – a rhetoric inherent in the postindustrial thesis of the 1960s, in the ideas of 'new economy' disciples in the 1990s, and in the radical and conservative paeans to the sharing economy of today – we still remain bound to a system of competition and profitability. Platforms offer new forms of competition and control, but in the end profitability is the great arbiter of success. Given these constraints, we must now open platforms up onto the wider economy. We can begin by returning to the scene of the long downturn and the problem of global manufacturing overcapacity. If we look to the US manufacturing sector, we find few signs that the sector is improving. In terms of output, manufacturing growth has dipped from an annual growth rate of 2.1 per cent between 1999 and 2008 to a rate of 1.3 per